Task 6

AI-Based Hit Scoring / QSAR Prototype

Objective

The objective of this task was to develop a machine learning–based QSAR (Quantitative Structure–Activity Relationship) prototype to support early-stage **hit prioritization for the validated target EGFR**.

The goal was not to predict exact experimental potency values, but to construct a **robust prioritization model** capable of ranking compounds by likelihood of biological activity, thereby reducing experimental screening burden and accelerating lead identification.

Dataset and Data Preparation

Data Source

- Database: **ChEMBL**
- Target: **EGFR (human)**
- Bioactivity types: **IC_{50}, Ki, Kd**
- Records after quality filtering: ~19,000 compound–activity pairs

Public ChEMBL data was selected due to its curated experimental annotations and broad usage in industrial QSAR benchmarking.

Data Cleaning and Standardization

To improve learning stability and reduce experimental artifacts, the following preprocessing steps were applied:

- Retained only quantitative bioactivity measurements (IC_{50}, Ki, Kd)
- Removed compounds with missing SMILES or activity values
- Standardized all activity measurements to nanomolar (nM)
- Transformed activities into **pActivity ($-\log_{10}$ molar)** scale

This transformation compresses extreme values, improves numerical behavior, and allows comparison across heterogeneous assays.

Descriptor Generation

Molecular descriptors were computed using **RDKit**, converting SMILES representations into chemically meaningful numerical features.

Descriptors included:

- Molecular weight
- Lipophilicity (LogP)
- Topological polar surface area (TPSA)
- Hydrogen bond donors and acceptors
- Number of rotatable bonds

These descriptors capture key physicochemical drivers of kinase–ligand binding while maintaining interpretability.

Modeling Strategy

Regression vs Classification

Two modeling paradigms were evaluated:

- **Regression**: prediction of continuous pActivity values
- **Classification**: prediction of active vs inactive compounds

Although regression offers quantitative outputs, classification was selected as the final hit-scoring strategy due to the intrinsic nature of public bioactivity data.

Classification is more appropriate because:

- Bioactivity datasets are highly imbalanced (few actives, many inactives)
- Experimental values vary significantly across laboratories
- Early-stage drug discovery decisions are binary in nature (test vs deprioritize)

Thus, classification better reflects real-world screening logic.

Machine Learning Models Evaluated

The following algorithms were benchmarked:

- Logistic Regression

- Random Forest Classifier
- Gradient Boosting Classifier

Models were trained using stratified train–test splits to preserve class balance.

Model Performance

Regression Benchmarking

Model	RMSE	R^2
Linear Regression	1.30	0.16
Random Forest Regressor	1.08	0.42
Gradient Boosting Regressor	1.21	0.28

While nonlinear regressors improved performance, predictive accuracy remained limited due to assay heterogeneity and experimental noise.

Classification Performance

Model	ROC-AUC
Logistic Regression	0.76
Random Forest Classifier	**0.87**
Gradient Boosting Classifier	0.81

The Random Forest classifier demonstrated the strongest ability to discriminate active compounds from inactives.

Final Model Selection

Selected Model: Random Forest Classifier

Justification:

- Highest ROC-AUC score
- Robust handling of nonlinear chemical relationships
- Reduced sensitivity to noisy activity labels

- Strong performance under class imbalance
- Interpretable feature importance patterns

Hit Scoring and Prioritization

The selected model was retrained on the full dataset and used to compute **hit probability scores** for all compounds.

Each molecule was assigned:

- Experimental pActivity
- Binary activity class
- Predicted probability of being active

Compounds were ranked by hit probability, enabling prioritization of high-confidence candidates for experimental follow-up.

Discussion: Model Reliability and Data Bias

While the QSAR model demonstrates strong predictive performance, several fundamental limitations must be considered.

Applicability Domain Bias

The model can reliably predict only compounds that fall within the chemical space represented in the training data. Since most ChEMBL EGFR inhibitors are **Lipinski-compliant small molecules**, predictions for macrocycles, peptides, or natural products may be unreliable.

This limitation defines the **applicability domain** of the model.

Activity Cliffs

In medicinal chemistry, small structural modifications can produce large and unpredictable changes in bioactivity. These **activity cliffs** are difficult for machine learning models to capture and represent a known limitation of QSAR approaches.

As a result, neighboring molecules in chemical space may exhibit divergent biological behavior.

Decoy Bias

If inactive compounds differ substantially from actives in basic physicochemical properties (e.g., molecular weight or lipophilicity), the model may learn superficial patterns rather than true binding chemistry.

This phenomenon, known as **decoy bias**, can artificially inflate model performance metrics and must be carefully considered when interpreting results.

Overall Reliability Assessment

Strengths

- Robust discrimination of actives vs inactives
- Scalable to large virtual screening libraries
- Aligns with early-stage decision-making frameworks

Limitations

- Sensitive to chemical space coverage
- Impacted by assay heterogeneity
- Limited interpretability at atomic-level interactions

Therefore, model predictions should be interpreted as **prioritization guidance**, not definitive activity predictions.

Final Conclusion

This AI-based QSAR prototype demonstrates that machine learning can effectively support hit prioritization for EGFR by integrating molecular descriptors with experimental bioactivity data.

The Random Forest classification model achieved strong discriminatory performance (ROC-AUC ≈ 0.87) while providing biologically plausible structure–activity relationships. Although constrained by applicability domain, activity cliffs, and data bias, the framework offers a practical and scalable tool for reducing discovery risk in early-stage drug discovery.